CEE6501 — Written Assignment, Week 7

Assigned: 02/27/2026 (Week 7)

Due: 03/12/2026

Canvas Submission Link: https://gatech.instructure.com/courses/517856/assignments/2320540

Logistics

📄 Assignment Format

This is a **written assignment**, not a coding assignment.

- Complete the assignment **by hand or typed up**
- You may write:
  - in a markdown file
  - on paper
  - on a tablet
  - using a stylus or digital note-taking app
- After completing the assignment, **scan or export your work** and upload it to Canvas

You may use **any format you prefer**, but your submission **must be neat and legible*

📤 Submission Instructions

- Upload **one file** to Canvas
- Accepted formats: **PDF, PNG, or JPG**
- Make sure all pages are clearly visible and in the correct order

✅ Checklist Before Submitting

- [ ] Assignment completed by hand
- [ ] Writing is neat and legible
- [ ] Title block provided on each page
- [ ] All questions answered
- [ ] File uploaded successfully to Canvas

🤝 Collaboration / AI tools

You may discuss concepts with classmates and you may use AI tools to help you learn, but **your submitted work must be written by you and you must understand it**. If you used outside help, add a short note in the final reflection cell.

Question 1 — Indeterminate Beam (DSM by Hand)

Using the methods from **Lecture 6**, solve the indeterminate beam below **by hand** using the **Direct Stiffness Method (DSM)**. Follow the step-by-step procedure at the end of **Lecture 6.3**.

Hand-written solutions are acceptable — but they must be **neat, clearly organized, and easy to follow**. Clearly label all steps, matrices, and intermediate results.

If possible, submit a clean solution in **Markdown** or a **Jupyter notebook**. A notebook allows you to structure your work clearly, embed images, and present your solution as a short technical report that walks through your full reasoning step by step.

Submit / report:

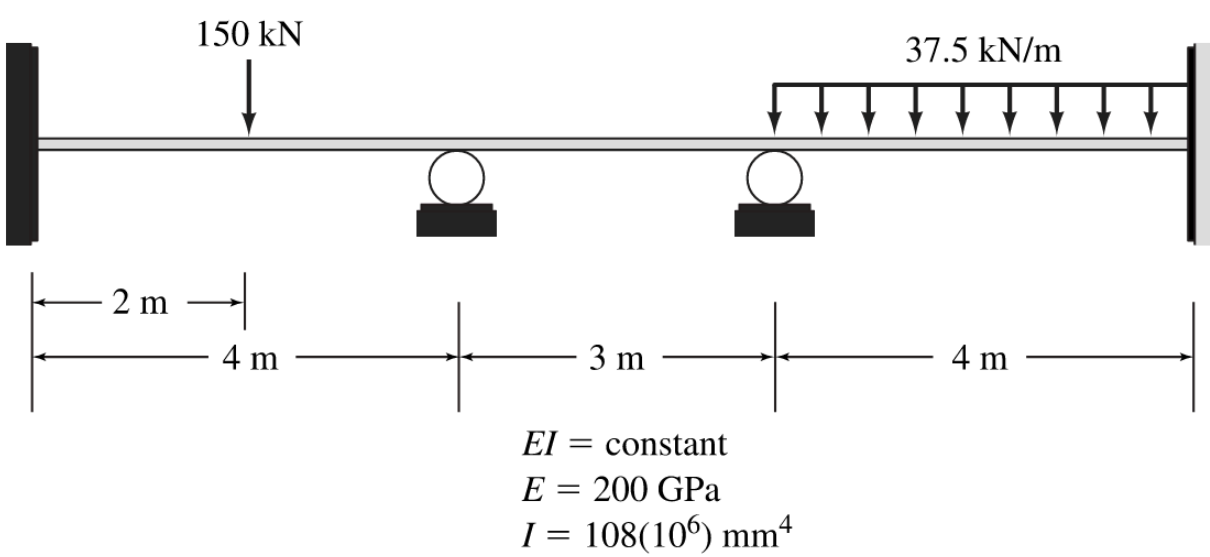
- Global displacements at the **free DOFs**
- Global reactions at the **restrained DOFs**
- A **global equilibrium check**
- Local element end forces
- Local element end displacements
- A clear, neat sketch of the **deformed shape**

Numbering conventions:

- Number global nodes **left → right**
- Use the Lecture 6 DOF convention (1-based):
  - $\mathrm{DOF}(j, u_y) = 2j - 1$
  - $\mathrm{DOF}(j, \theta) = 2j$

Structural info:

- E, I are constant
- $E = 200\ \mathrm{GPa}$
- $I = 108 \times 10^6\ \mathrm{mm}^4$



Solution:

Node numbering and coordinates

- Node 1: x=0 m (fixed)
- Node 2: x=4 m (roller)
- Node 3: x=7 m (roller)
- Node 4: x=11 m (fixed)

Global DOF numbering

- Node 1:

$$u_1 = u_{y1}, \quad u_2 = \theta_1$$

- Node 2:

$$u_3 = u_{y2}, \quad u_4 = \theta_2$$

- Node 3:

$$u_5 = u_{y3}, \quad u_6 = \theta_3$$

- Node 4:

$$u_7 = u_{y4}, \quad u_8 = \theta_4$$

Element connectivity

- Element 1 (1-2): e1
- Element 2 (2-3): e2
- Element 3 (3-4): e3

Boundary conditions

- Free DOFs:

$$u_4, u_6$$

- Restrained DOFs:

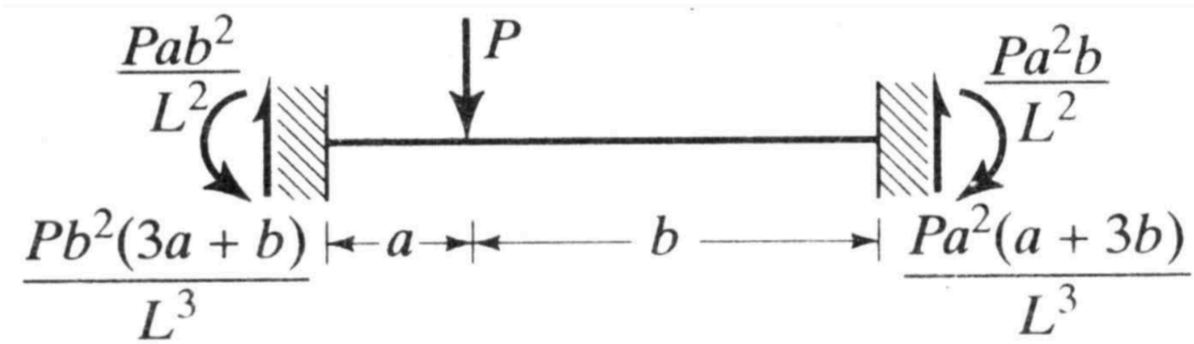
$$u_1, u_2, u_3, u_5, u_7, u_8 = 0$$

Therefore, DOF vector

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ u_4 \\ 0 \\ u_6 \\ 0 \\ 0 \end{bmatrix}$$

Applied loads and equivalent nodal loads

- **Point load at midspan of Element 1**

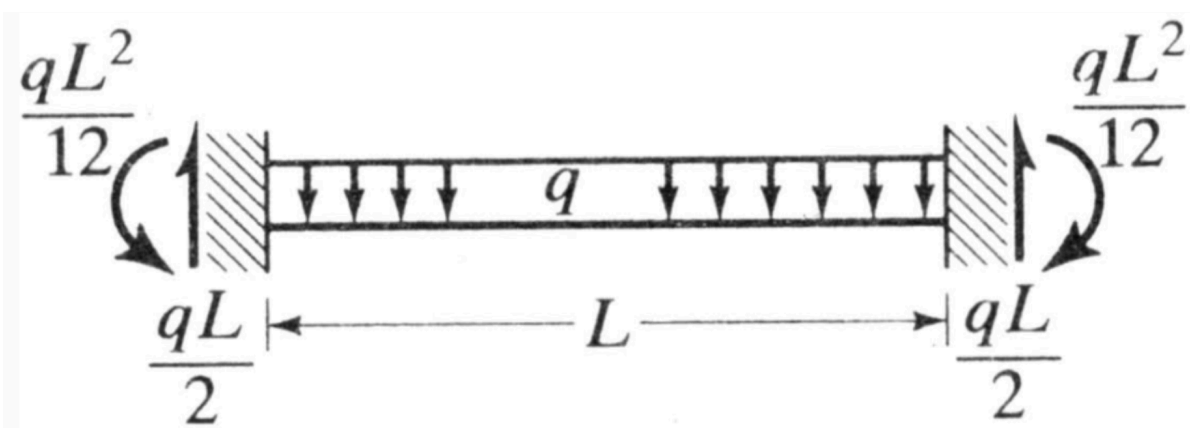


$$P = 150kN, \quad L_1 = 4, \quad a = b = 2$$

$$[u_1, u_2, u_3, u_4]$$

$$\mathbf{F}_{e1}^F = \begin{bmatrix} \frac{P}{2} \\ \frac{PL_1}{8} \\ \frac{P}{2} \\ -\frac{PL_1}{8} \end{bmatrix} = \begin{bmatrix} 75 \\ 75 \\ 75 \\ -75 \end{bmatrix}$$

- **Uniform load on Element 3**



$$q = 37.5kN/m, \quad L_3 = 4$$

$$[u_5, u_6, u_7, u_8]$$

$$\mathbf{F}_{e3}^{F} = \begin{bmatrix} \frac{qL_3}{2} \\ \frac{qL_3^2}{12} \\ \frac{qL_3}{2} \\ -\frac{qL_3^2}{12} \end{bmatrix} = \begin{bmatrix} 75 \\ 50 \\ 75 \\ -50 \end{bmatrix}$$

Axial Force vector

$$\mathbf{F} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Fixed End Force vector

$$\mathbf{F}^{F} = \mathbf{F}_{e1}^{F} + \mathbf{F}_{e3}^{F} = \begin{bmatrix} 75 \\ 75 \\ 75 \\ -75 \\ 75 \\ 50 \\ 75 \\ -50 \end{bmatrix}$$

Element-to-Global Mapping

4x4 Beam Element Stiffness Matrix k

$$\mathbf{k}_{ij,kl} = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 24 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

DOF order per element:

$$[u_i, u_j, u_k, u_l]$$

ELement 1 and 3 (L=4000 mm):

$$I = 108 \times 10^6 \text{ mm}^4 = 1.08 \times 10^{-4} \text{ m}^4$$

$$EI = (200 \times 10^9)(1.08 \times 10^{-4}) = 2.16 \times 10^7 N \cdot m^2 = 21600 kN \cdot m^2$$

$$\mathbf{k}_{12,34} = \mathbf{k}_{56,78} = \begin{bmatrix} 4050 & 8100 & -4050 & 8100 \\ 8100 & 21600 & -8100 & 10800 \\ -4050 & -8100 & 4050 & -8100 \\ 8100 & 10800 & -8100 & 21600 \end{bmatrix}$$

Element 2 (L=3000mm):

$$\mathbf{k}_{34,56} = \begin{bmatrix} 9600 & 14400 & -9600 & 14400 \\ 14400 & 28800 & -14400 & 14400 \\ -9600 & -14400 & 9600 & -14400 \\ 14400 & 14400 & -14400 & 28800 \end{bmatrix}$$

Global Stiffness Matrix K

$$\mathbf{k}_{12,34}, \mathbf{k}_{34,56}, \mathbf{k}_{56,78} \rightarrow \mathbf{K}$$

$$\mathbf{K} = \begin{bmatrix} 4050 & 8100 & -4050 & 8100 & 0 & 0 & 0 & 0 \\ 8100 & 21600 & -8100 & 10800 & 0 & 0 & 0 & 0 \\ -4050 & -8100 & 13650 & 6300 & -9600 & 14400 & 0 & 0 \\ 8100 & 10800 & 6300 & 50400 & -14400 & 14400 & 0 & 0 \\ 0 & 0 & -9600 & -14400 & 13650 & -6300 & -4050 & 8100 \\ 0 & 0 & 14400 & 14400 & -6300 & 50400 & -8100 & 10800 \\ 0 & 0 & 0 & 0 & -4050 & -8100 & 4050 & -8100 \\ 0 & 0 & 0 & 0 & 8100 & 10800 & -8100 & 21600 \end{bmatrix}$$

Partitioned Matrix Form

$$\begin{bmatrix} \mathbf{K}_{ff} & \mathbf{K}_{fr} \\ \mathbf{K}_{rf} & \mathbf{K}_{rr} \end{bmatrix} \begin{bmatrix} \mathbf{u}_f \\ \mathbf{u}_r \end{bmatrix} = \begin{bmatrix} \mathbf{F}_f \\ \mathbf{F}_r \end{bmatrix} - \begin{bmatrix} \mathbf{F}_f^F \\ \mathbf{F}_r^F \end{bmatrix}$$

Reorder the stiffness matrix

$$\mathbf{K}_{new} = \mathbf{K}([4,6,1,2,3,5,7,8],[4,6,1,2,3,5,7,8])$$

$$= \begin{bmatrix} 50400 & 14400 & 8100 & 10800 & 6300 & -14400 & 0 & 0 \\ 14400 & 50400 & 0 & 0 & 14400 & -6300 & -8100 & 10800 \\ 8100 & 0 & 4050 & 8100 & -4050 & 0 & 0 & 0 \\ 10800 & 0 & 8100 & 21600 & -8100 & 0 & 0 & 0 \\ 6300 & 14400 & -4050 & -8100 & 13650 & -9600 & 0 & 0 \\ -14400 & -6300 & 0 & 0 & -9600 & 13650 & -4050 & 8100 \\ 0 & -8100 & 0 & 0 & 0 & -4050 & 4050 & -8100 \\ 0 & 10800 & 0 & 0 & 0 & 8100 & -8100 & 21600 \end{bmatrix}$$

$$= \begin{bmatrix} \mathbf{K}_{ff} & \mathbf{K}_{fr} \\ \mathbf{K}_{rf} & \mathbf{K}_{rr} \end{bmatrix}$$

Therefore,

$$\mathbf{K}_{ff} = \begin{bmatrix} 50400 & 14400 \\ 14400 & 50400 \end{bmatrix}$$

Solve Global displacements at the free DOFs

$$\mathbf{u}_f = \begin{bmatrix} u_4 \\ u_6 \end{bmatrix}, \qquad \mathbf{u}_r = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{F}_f = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \mathbf{F}_r = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{F}_f^F = \begin{bmatrix} -75 \\ 50 \end{bmatrix}, \qquad \mathbf{F}_r^F = \begin{bmatrix} 75 \\ 75 \\ 75 \\ 75 \\ 75 \\ -50 \end{bmatrix}$$

$$\mathbf{K}_{ff}\mathbf{u}_f + \mathbf{K}_{fr}\mathbf{u}_r = \mathbf{F}_f - \mathbf{F}_f^F \qquad and \qquad \mathbf{u}_r = 0$$

$$\rightarrow \qquad \mathbf{K}_{ff}\mathbf{u}_f = \mathbf{F}_f - \mathbf{F}_f^F \qquad \rightarrow \qquad \mathbf{u}_f = \mathbf{K}_{ff}^{-1}(\mathbf{F}_f - \mathbf{F}_f^F) = \begin{bmatrix} 50400 & 14400 \\ 14400 & 50400 \end{bmatrix}^{-1} \begin{bmatrix} 75 \\ -50 \end{bmatrix}$$

Therefore,

$$u_4 = \theta_2 = 1.929 \times 10^{-3} \text{ rad}$$

$$u_6 = \theta_3 = -1.543 \times 10^{-3} \text{ rad}$$

Compute Global reactions at the restrained DOFs

$$\mathbf{F}_r = \mathbf{K}_{rf}\mathbf{u}_f + \mathbf{K}_{rr}\mathbf{u}_r + \mathbf{F}_r^F$$

$$= \begin{bmatrix} 8100 & 0 \\ 10800 & 0 \\ 6300 & 14400 \\ -14400 & -6300 \\ 0 & -8100 \\ 0 & 10800 \end{bmatrix} \begin{bmatrix} 0.00192901 \\ -0.00154321 \end{bmatrix} + \begin{bmatrix} 75 \\ 75 \\ 75 \\ 75 \\ 75 \\ -50 \end{bmatrix} = \begin{bmatrix} 90.625 \\ 95.833 \\ 64.931 \\ 56.917 \\ 87.500 \\ -66.667 \end{bmatrix}$$

Reactions:

- Node 1:

$$R_1 = 90.625kN, \qquad R_2 = 95.833kN \cdot m$$

- Node 2:

$$R_3 = 64.931kN$$

- Node 3:

$$R_5 = 56.944kN$$

- Node 4:

$$R_7 = 87.500kN, \qquad R_8 = -66.667kN \cdot m \qquad (counterclockwise)$$

Global equilibrium check

Vertical force equilibrium External loads:

$$P = -150 \text{ kN}, \qquad q = -37.5 \text{ kN/m}, \qquad L_3 = 4 \text{ m}$$

$$\sum F_y^{ext} = P + qL_3 = (-150) + (-37.5)(4) = -300 \text{ kN}$$

$$R_{y1} = 90.625, \quad R_3 = 64.931, \quad R_5 = 56.944, \quad R_7 = 87.500 \text{ kN}$$

$$\sum R_y = 90.625 + 64.931 + 56.944 + 87.500 = 300.000 \text{ kN}$$

$$\sum R_y + \sum F_y^{ext} = 300.000 - 300.000 = 0$$

Moment equilibrium about Node 1

$$M_P = P(2) = (-150)(2) = -300\ kN\cdot m$$

$$M_q = (qL_3)(9) = (-150)(9) = -1350\ kN\cdot m$$

$$\sum M_{x=0}^{ext} = M_P + M_q = -300 - 1350 = -1650\ kN\cdot m$$

$$M_1 = R_2 = 95.833, \qquad M_4 = R_8 = -66.667\ kN\cdot m$$

$$\sum M_{x=0}^{react} = M_1 + M_4 + R_3(4) + R_5(7) + R_7(11)$$

$$\sum M_{x=0}^{react} = 95.833 - 66.667 + 64.931(4) + 56.944(7) + 87.500(11) = 1650.000\ kN\cdot m$$

$$\sum M_{x=0}^{react} + \sum M_{x=0}^{ext} = 1650.000 - 1650.000 = 0$$

Local element end displacements

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.00192901 \\ 0 \\ -0.00154321 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Element DOF order: } [u_{yi}, \theta_i, u_{yj}, \theta_j]$$

$$\mathbf{u}_{e1} = \begin{bmatrix} u_{y1} \\ \theta_1 \\ u_{y2} \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.00192901 \end{bmatrix} \qquad (1\text{–}2)$$

$$\mathbf{u}_{e2} = \begin{bmatrix} u_{y2} \\ \theta_2 \\ u_{y3} \\ \theta_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.00192901 \\ 0 \\ -0.00154321 \end{bmatrix} \qquad (2\text{–}3)$$

$$\mathbf{u}_{e3} = \begin{bmatrix} u_{y3} \\ \theta_3 \\ u_{y4} \\ \theta_4 \end{bmatrix} = \begin{bmatrix} 0 \\ -0.00154321 \\ 0 \\ 0 \end{bmatrix} \qquad (3\text{–}4)$$

Local element and forces

$$\text{End force order: } [V_i, M_i, V_j, M_j]$$

$$\mathbf{Q} = \mathbf{k}_e \mathbf{u}_e + \mathbf{Q}_f^{(e)}$$

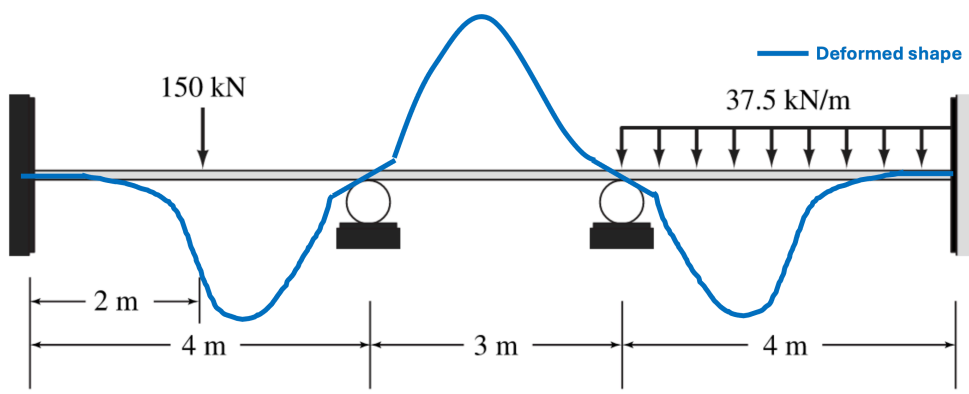
$$\mathbf{Q}^{(e1)} = \begin{bmatrix} 4050 & 8100 & -4050 & 8100 \\ 8100 & 21600 & -8100 & 10800 \\ -4050 & -8100 & 4050 & -8100 \\ 8100 & 10800 & -8100 & 21600 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.00192901 \end{bmatrix} + \begin{bmatrix} 75 \\ 75 \\ 75 \\ -75 \end{bmatrix} = \begin{bmatrix} 90.625 \\ 95.833 \\ 59.375 \\ -33.333 \end{bmatrix} \; kN, kN \cdot m$$

$$\mathbf{Q}^{(e2)} = \begin{bmatrix} 9600 & 14400 & -9600 & 14400 \\ 14400 & 28800 & -14400 & 14400 \\ -9600 & -14400 & 9600 & -14400 \\ 14400 & 14400 & -14400 & 28800 \end{bmatrix} \begin{bmatrix} 0 \\ 0.00192901 \\ 0 \\ -0.00154321 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 5.556 \\ 33.333 \\ -5.556 \\ -16.667 \end{bmatrix} \; kN, kN \cdot m$$

$$\mathbf{Q}^{(e3)} = \begin{bmatrix} 4050 & 8100 & -4050 & 8100 \\ 8100 & 21600 & -8100 & 10800 \\ -4050 & -8100 & 4050 & -8100 \\ 8100 & 10800 & -8100 & 21600 \end{bmatrix} \begin{bmatrix} 0 \\ -0.00154321 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 75 \\ 50 \\ 75 \\ -50 \end{bmatrix} = \begin{bmatrix} 62.500 \\ 16.667 \\ 87.500 \\ -66.667 \end{bmatrix} \; kN, kN \cdot m$$

Deformed shape

- Sketch a smooth deflection curve that bends upward/downward between spans consistent with these signs, returning to zero deflection at each support.



$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.00192901 \\ 0 \\ -0.00154321 \\ 0 \\ 0 \end{bmatrix}$$

Question 2 — Indeterminate Frame (DSM by Hand)

Using the methods from **Lecture 7**, solve the indeterminate frame below **by hand** using the **Direct Stiffness Method (DSM)**. Follow the step-by-step procedure at the end of **Lecture 7.2**.

Hand-written solutions are acceptable — but they must be **neat, clearly organized, and easy to follow**. Clearly label all steps, matrices, and intermediate results.

If possible, submit a clean solution in **Markdown** or a **Jupyter notebook**. A notebook allows you to structure your work clearly, embed images, and present your solution as a short technical report that walks through your full reasoning step by step.

Submit / report:

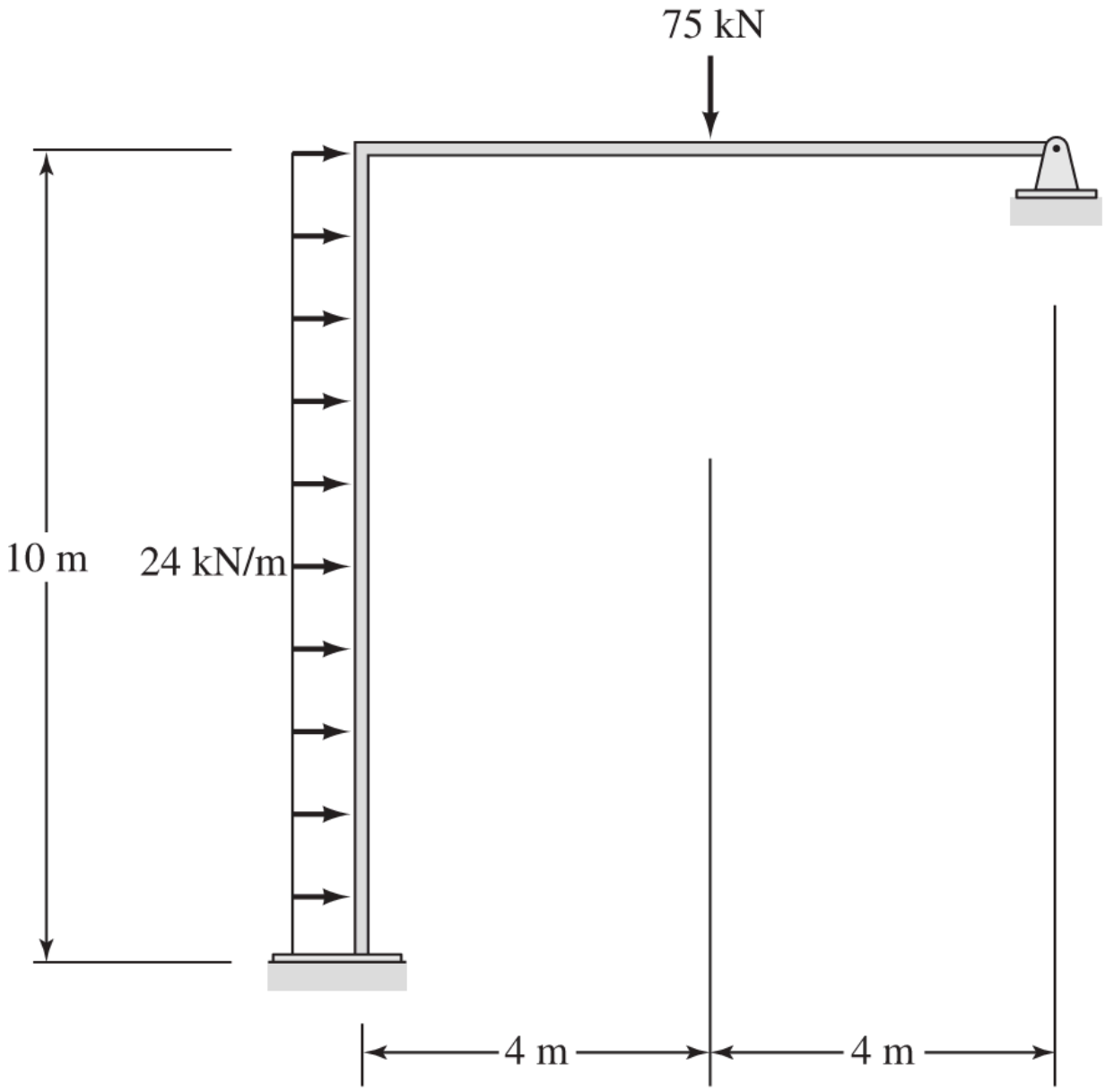
- Global displacements at the **free DOFs**
- Global reactions at the **restrained DOFs**
- A **global equilibrium check**
- Local element end forces
- Local element end displacements
- A clear, neat sketch of the **deformed shape**

Numbering conventions:

- Number global nodes **bottom-left → top-left → top-right**
- Use the Lecture 7 DOF convention (1-based):
  - $\mathrm{DOF}(j, u_x) = 3j - 2$
  - $\mathrm{DOF}(j, u_y) = 3j - 1$
  - $\mathrm{DOF}(j, \theta) = 3j$

Structural info:

- E, A, I are constant
- $E = 200\ \mathrm{GPa}$
- $A = 4740\ \mathrm{mm}^2$
- $I = 22.2 \times 10^6\ \mathrm{mm}^4$


75 kN
10 m
24 kN/m
4 m
4 m

Solution:

Node numbering and coordinates

- Node 1: x=0 m, y=0 m (fixed)
- Node 2: x=0 m, y=10 m
- Node 3: x=8 m, y=10 m (pin)

Global DOF numbering

- Node 1:

$$u_1 = u_{x1}, \quad u_2 = u_{y1}, \quad u_3 = \theta_1$$

- Node 2:

$$u_4 = u_{x2}, \quad u_5 = u_{y2}, \quad u_6 = \theta_2$$

- Node 3:

$$u_7 = u_{x2}, \quad u_8 = u_{y3}, \quad u_9 = \theta_3$$

Element connectivity

- Element 1 (1-2): e1
- Element 2 (2-3): e2

Boundary conditions

- Free DOFs:

$$u_4, u_5, u_6, u_9$$

- Restrained DOFs:

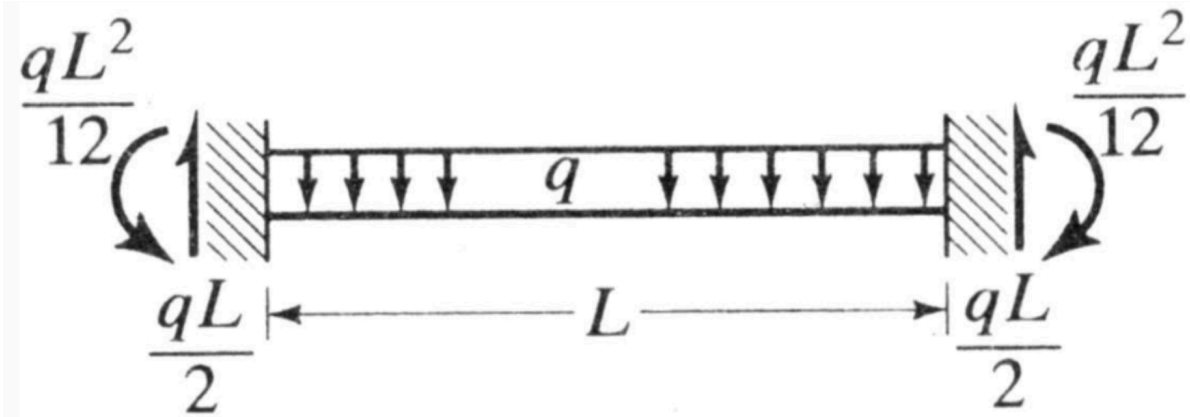
$$u_1, u_2, u_3, u_7, u_8 = 0$$

Therefore, DOF vector

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \\ u_9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ u_4 \\ u_5 \\ u_6 \\ 0 \\ 0 \\ u_9 \end{bmatrix}$$

Applied loads and equivalent nodal loads

- **Uniform load on Element 1**

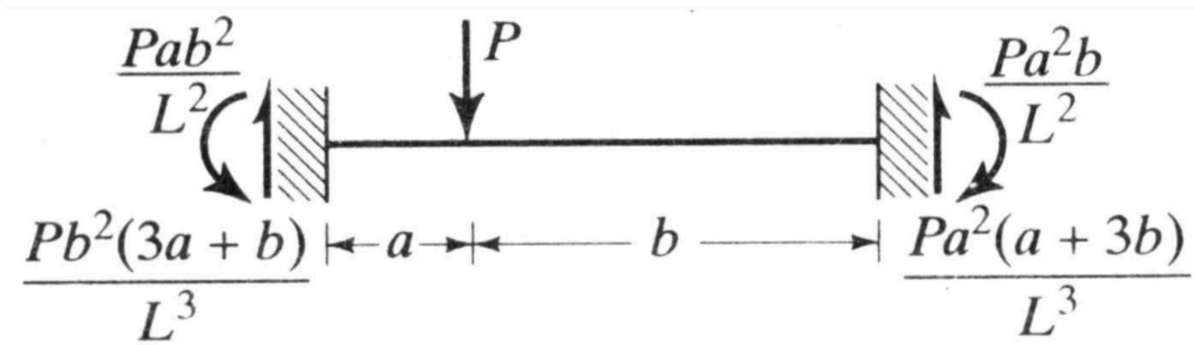


$$q = 24\ kN/m, \quad L = 10\ m$$

$$[u_{x1}, u_{y1}, \theta_1, u_{x2}, u_{y2}, \theta_2]$$

$$\mathbf{F}_{e1}^F = \begin{bmatrix} -\frac{qL}{2} \\ 0 \\ \frac{qL^2}{12} \\ -\frac{qL}{2} \\ 0 \\ -\frac{qL^2}{12} \end{bmatrix} = \begin{bmatrix} -120 \\ 0 \\ 200 \\ -120 \\ 0 \\ -200 \end{bmatrix} \quad (kN, kN\cdot m)$$

- **Point load at midspan of Element 2**



$$P = 75kN, \quad L = 8, \quad a = b = 4$$

$$[u_{x2}, u_{y2}, \theta_2, u_{x3}, u_{y3}, \theta_3]$$

$$\mathbf{F}_{e2}^F = \begin{bmatrix} 0 \\ \frac{P}{2} \\ \frac{PL}{8} \\ 0 \\ \frac{P}{2} \\ -\frac{PL}{8} \end{bmatrix} = \begin{bmatrix} 0 \\ 37.5 \\ 75 \\ 0 \\ 37.5 \\ -75 \end{bmatrix} \quad (kN, kN\cdot m)$$

Therefore,

Axial Force vector

$$\mathbf{F} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Fixed End Force vector

$$\mathbf{F}^F = \mathbf{F}_{e1}^F + \mathbf{F}_{e2}^F = \begin{bmatrix} -120 \\ 0 \\ 200 \\ -120 \\ 37.5 \\ -125 \\ 0 \\ 37.5 \\ -75 \end{bmatrix}$$

Element-to-Global Mapping and stiffness matrices

2D frame element stiffness matrix (local) DOF order:

$$[u_{xi}, u_{yi}, \theta_i, u_{xj}, u_{yj}, \theta_j]$$

Local stiffness:

$$\mathbf{k}_e = \begin{bmatrix} \frac{EA}{L} & 0 & 0 & -\frac{EA}{L} & 0 & 0 \\ 0 & \frac{12EI}{L^3} & \frac{6EI}{L^2} & 0 & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{4EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{2EI}{L} \\ -\frac{EA}{L} & 0 & 0 & \frac{EA}{L} & 0 & 0 \\ 0 & -\frac{12EI}{L^3} & -\frac{6EI}{L^2} & 0 & \frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{2EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{4EI}{L}[4pt] \end{bmatrix}$$

Element lengths

$$L_{e1} = 10\ \text{m}, \qquad L_{e2} = 8\ \text{m}$$

Compute global stiffness for each element

Let

$$c = \cos\theta, \qquad s = \sin\theta$$

Then transformation matrix is

$$\mathbf{T} = \begin{bmatrix} c & s & 0 & 0 & 0 & 0 \\ -s & c & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & c & s & 0 \\ 0 & 0 & 0 & -s & c & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

So the **global** element stiffness is

$$\mathbf{k}_e = \mathbf{T}^T \mathbf{k}_e' \mathbf{T}$$

ELement 1:

$$I = 22.2 \times 10^6\ \text{mm}^4 = 0.222 \times 10^{-4}\ \text{m}^4$$

$$EI = (200 \times 10^9)(0.222 \times 10^{-4}) = 44.4 \times 10^5 N \cdot m^2 = 4440 kN \cdot m^2$$

$$A = 4740\ \text{mm}^2, \qquad L = 10\ m$$

$$\frac{EA}{L} = 94800, \quad \frac{12EI}{L^3} = 53.28, \quad \frac{6EI}{L^2} = 266.4, \quad \frac{4EI}{L} = 1776, \quad \frac{2EI}{L} = 888$$

$$\mathbf{k}_{e1}' = \begin{bmatrix} 94800 & 0 & 0 & -94800 & 0 & 0 \\ 0 & 53.28 & 266.4 & 0 & -53.28 & 266.4 \\ 0 & 266.4 & 1776 & 0 & -266.4 & 888 \\ -94800 & 0 & 0 & 94800 & 0 & 0 \\ 0 & -53.28 & -266.4 & 0 & 53.28 & -266.4 \\ 0 & 266.4 & 888 & 0 & -266.4 & 1776 \end{bmatrix}$$

$$\theta = 90^\circ \Rightarrow c = 0,\ s = 1$$

$$\mathbf{T}_{e1} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{k}_{e1} = \mathbf{T}^T \mathbf{k}'_{e1} \mathbf{T} = \begin{bmatrix} 53.28 & 0 & -266.4 & -53.28 & 0 & -266.4 \\ 0 & 94800 & 0 & 0 & -94800 & 0 \\ -266.4 & 0 & 1776 & 266.4 & 0 & 888 \\ -53.28 & 0 & 266.4 & 53.28 & 0 & 266.4 \\ 0 & -94800 & 0 & 0 & 94800 & 0 \\ -266.4 & 0 & 888 & 266.4 & 0 & 1776 \end{bmatrix}$$

Element 2:

$$I = 22.2 \times 10^6 \text{ mm}^4 = 0.222 \times 10^{-4} \text{ m}^4$$

$$EI = (200 \times 10^9)(0.222 \times 10^{-4}) = 44.4 \times 10^5 N \cdot m^2 = 4440 kN \cdot m^2$$

$$A = 4740 \text{ mm}^2, \qquad L = 8\ m$$

No transformation needed.

$$\mathbf{k}^{e2} = \mathbf{k}'_{e2} = \begin{bmatrix} 118500 & 0 & 0 & -118500 & 0 & 0 \\ 0 & 104.0625 & 416.25 & 0 & -104.0625 & 416.25 \\ 0 & 416.25 & 2220 & 0 & -416.25 & 1110 \\ -118500 & 0 & 0 & 118500 & 0 & 0 \\ 0 & -104.0625 & -416.25 & 0 & 104.0625 & -416.25 \\ 0 & 416.25 & 1110 & 0 & -416.25 & 2220 \end{bmatrix}$$

Global Stiffness Matrix K

$$\mathbf{k}_{e1}, \mathbf{k}_{e2} \rightarrow \mathbf{K}$$

$$\mathbf{K} = \begin{bmatrix} 53.28 & 0 & -266.4 & -53.28 & 0 & -266.4 & 0 & 0 & 0 \\ 0 & 94800 & 0 & 0 & -94800 & 0 & 0 & 0 & 0 \\ -266.4 & 0 & 1776 & 266.4 & 0 & 888 & 0 & 0 & 0 \\ -53.28 & 0 & 266.4 & 118553.28 & 0 & 266.4 & -118500 & 0 & 0 \\ 0 & -94800 & 0 & 0 & 94904.0625 & 416.25 & 0 & -104.0625 & 416.25 \\ -266.4 & 0 & 888 & 266.4 & 416.25 & 3996 & 0 & -416.25 & 1110 \\ 0 & 0 & 0 & -118500 & 0 & 0 & 118500 & 0 & 0 \\ 0 & 0 & 0 & 0 & -104.0625 & -416.25 & 0 & 104.0625 & -416.25 \\ 0 & 0 & 0 & 0 & 416.25 & 1110 & 0 & -416.25 & 2220 \end{bmatrix}$$

Partitioned Matrix Form

$$\begin{bmatrix} \mathbf{K}_{ff} & \mathbf{K}_{fr} \\ \mathbf{K}_{rf} & \mathbf{K}_{rr} \end{bmatrix} \begin{bmatrix} \mathbf{u}_f \\ \mathbf{u}_r \end{bmatrix} = \begin{bmatrix} \mathbf{F}_f \\ \mathbf{F}_r \end{bmatrix} - \begin{bmatrix} \mathbf{F}_f^F \\ \mathbf{F}_r^F \end{bmatrix}$$

Reorder the stiffness matrix

$$\mathbf{K}_{new} = \mathbf{K}([4,5,6,9,1,2,3,7,8],[4,5,6,9,1,2,3,7,8])$$

$$= \begin{bmatrix} 118553.28 & 0 & 266.4 & 0 & -53.28 & 0 & 266.4 & -118500 & 0 \\ 0 & 94904.0625 & 416.25 & 416.25 & 0 & -94800 & 0 & 0 & -104.0625 \\ 266.4 & 416.25 & 3996 & 1110 & -266.4 & 0 & 888 & 0 & -416.25 \\ 0 & 416.25 & 1110 & 2220 & 0 & 0 & 0 & 0 & -416.25 \\ -53.28 & 0 & -266.4 & 0 & 53.28 & 0 & -266.4 & 0 & 0 \\ 0 & -94800 & 0 & 0 & 0 & 94800 & 0 & 0 & 0 \\ 266.4 & 0 & 888 & 0 & -266.4 & 0 & 1776 & 0 & 0 \\ -118500 & 0 & 0 & 0 & 0 & 0 & 0 & 118500 & 0 \\ 0 & -104.0625 & -416.25 & -416.25 & 0 & 0 & 0 & 0 & 104.0625 \end{bmatrix}$$

$$= \begin{bmatrix} \mathbf{K}_{ff} & \mathbf{K}_{fr} \\ \mathbf{K}_{rf} & \mathbf{K}_{rr} \end{bmatrix}$$

Therefore,

$$\mathbf{K}_{ff} = \begin{bmatrix} 118553.28 & 0 & 266.4 & 0 \\ 0 & 94904.0625 & 416.25 & 416.25 \\ 266.4 & 416.25 & 3996 & 1110 \\ 0 & 416.25 & 1110 & 2220 \end{bmatrix}$$

Solve Global displacements at the free DOFs

$$\mathbf{u}_f = \begin{bmatrix} u_4 \\ u_5 \\ u_6 \\ u_9 \end{bmatrix}, \qquad \mathbf{u}_r = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{F}_f = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad \mathbf{F}_r = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{F}_f^F = \begin{bmatrix} -120 \\ 37.5 \\ -125 \\ -75 \end{bmatrix}, \qquad \mathbf{F}_r^F = \begin{bmatrix} -120 \\ 0 \\ 200 \\ 0 \\ 37.5 \end{bmatrix}$$

$$\mathbf{K}_{ff}\mathbf{u}_f + \mathbf{K}_{fr}\mathbf{u}_r = \mathbf{F}_f - \mathbf{F}_f^F \qquad \text{and} \qquad \mathbf{u}_r = \mathbf{0}$$

$$\Rightarrow \quad \mathbf{K}_{ff}\mathbf{u}_f = \mathbf{F}_f - \mathbf{F}_f^F \quad \Rightarrow \quad \mathbf{u}_f = \mathbf{K}_{ff}^{-1}(\mathbf{F}_f - \mathbf{F}_f^F)$$

Using

$$\mathbf{K}_{ff} = \begin{bmatrix} 118553.28 & 0 & 266.4 & 0 \\ 0 & 94904.0625 & 416.25 & 416.25 \\ 266.4 & 416.25 & 3996 & 1110 \\ 0 & 416.25 & 1110 & 2220 \end{bmatrix}$$

$$\mathbf{F}_f - \mathbf{F}_f^F = -\begin{bmatrix} -120 \\ 37.5 \\ -125 \\ -75 \end{bmatrix} = \begin{bmatrix} 120 \\ -37.5 \\ 125 \\ 75 \end{bmatrix}$$

So

$$\begin{bmatrix} u_4 \\ u_5 \\ u_6 \\ u_9 \end{bmatrix} = \begin{bmatrix} 118553.28 & 0 & 266.4 & 0 \\ 0 & 94904.0625 & 416.25 & 416.25 \\ 266.4 & 416.25 & 3996 & 1110 \\ 0 & 416.25 & 1110 & 2220 \end{bmatrix}^{-1} \begin{bmatrix} 120 \\ -37.5 \\ 125 \\ 75 \end{bmatrix}$$

Therefore,

$$u_4 = u_{x2} = 9.5515 \times 10^{-4} \text{ m}$$

$$u_5 = u_{y2} = -5.9949 \times 10^{-4} \text{ m}$$

$$u_6 = \theta_2 = 2.5391 \times 10^{-2} \text{ rad}$$

$$u_9 = \theta_3 = 2.1201 \times 10^{-2} \text{ rad}$$

Compute Global reactions at the restrained DOFs

$$\mathbf{F}_r = \mathbf{K}_{rf}\mathbf{u}_f + \mathbf{K}_{rr}\mathbf{u}_r + \mathbf{F}_r^F \qquad \text{and} \qquad \mathbf{u}_r = \mathbf{0}$$

$$\Rightarrow \qquad \mathbf{F}_r = \mathbf{K}_{rf}\mathbf{u}_f + \mathbf{F}_r^F$$

$$\mathbf{u}_f = \begin{bmatrix} u_4 \\ u_5 \\ u_6 \\ u_9 \end{bmatrix} = \begin{bmatrix} 0.00095515 \\ -0.00059949 \\ 0.02539097 \\ 0.02120070 \end{bmatrix}$$

$$\mathbf{F}_r^F = \begin{bmatrix} -120 \\ 0 \\ 200 \\ 0 \\ 37.5 \end{bmatrix}$$

$$\mathbf{F}_r = \begin{bmatrix} -126.815 \\ 56.831 \\ 222.802 \\ -113.185 \\ 18.169 \end{bmatrix} \qquad (kN, kN, kN\cdot m, kN, kN)$$

Reactions:

- Node 1:

$$R_{x1} = -126.815 \text{ kN}, \qquad R_{y1} = 56.831 \text{ kN}, \qquad M_1 = 222.802\ kN\cdot m$$

- Node 3:

$$R_{x3} = -113.185 \text{ kN}, \qquad R_{y3} = 18.169 \text{ kN}$$

Global equilibrium check

Vertical force equilibrium ((+y) upward)

External loads:

$$P = -75 \text{ kN}, \qquad w = 24 \text{ kN/m}, \qquad L_{12} = 10 \text{ m}$$

Total vertical external force:

$$\sum F_y^{ext} = P = -75 \text{ kN}$$

Sum of vertical reactions:

$$\sum R_y = R_{y1} + R_{y3} = 56.831 + 18.169 = 75.000 \text{ kN}$$

Therefore,

$$\sum R_y + \sum F_y^{ext} = 75.000 - 75.000 = 0$$

Horizontal force equilibrium ((+x) to the right)

Column horizontal UDL resultant:

$$W = wL_{12} = 24(10) = 240 \text{ kN}$$

Sum of horizontal reactions:

$$\sum R_x = R_{x1} + R_{x3} = -126.815 - 113.185 = -240.000 \text{ kN}$$

Therefore,

$$\sum R_x + \sum F_x^{ext} = -240.000 + 240.000 = 0$$

Moment equilibrium about Node 1 (CCW (+))

External moments about Node 1:

- Column UDL resultant 240 kN at (y=5) m:

$$M_w = -240(5) = -1200 \; kN \cdot m$$

- Beam point load at (x=4) m:

$$M_P = -75(4) = -300 \; kN \cdot m$$

$$\sum M_{x=0}^{ext} = M_w + M_P = -1200 - 300 = -1500 \; kN \cdot m$$

Reaction moments about Node 1:

- Fixed-end reaction moment at Node 1:

$$M_1 = 222.802\ kN\cdot m$$

- Reactions at Node 3 located at ((x,y)=(8,10)):

$$M_{(R_{x3},R_{y3})} = xR_{y3} - yR_{x3} = 8(18.169) - 10(-113.185) = 1277.198\ kN\cdot m$$

So,

$$\sum M_{x=0}^{react} = M_1 + \left(xR_{y3} - yR_{x3}\right) = 222.802 + 1277.198 = 1500.000\ kN\cdot m$$

Therefore,

$$\sum M_{x=0}^{react} + \sum M_{x=0}^{ext} = 1500.000 - 1500.000 = 0$$

Local element end displacements

For 2D frame element, local DOF order:

$$[u'_{xi}, u'_{yi}, \theta_i,\ u'_{xj}, u'_{yj}, \theta_j]$$

Global displacement vector:

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \\ u_9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.00095515 \\ -0.00059949 \\ 0.02539097 \\ 0 \\ 0 \\ 0.02120070 \end{bmatrix}$$

Element 1:

$$\theta = 90^\circ \Rightarrow c = 0,\ s = 1$$

$$u'_x = cu_x + su_y = u_y, \qquad u'_y = -su_x + cu_y = -u_x$$

So the local element end displacement vector is

$$\mathbf{u}'_{e1} = \begin{bmatrix} u'_{x1} \\ u'_{y1} \\ \theta_1 \\ u'_{x2} \\ u'_{y2} \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ u_{y2} \\ -u_{x2} \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -0.00059949 \\ -0.00095515 \\ 0.02539097 \end{bmatrix}$$

Element 2: No translation needed.

$$\mathbf{u}'_{e2} = \begin{bmatrix} u_{x2} \\ u_{y2} \\ \theta_2 \\ u_{x3} \\ u_{y3} \\ \theta_3 \end{bmatrix} = \begin{bmatrix} 0.00095515 \\ -0.00059949 \\ 0.02539097 \\ 0 \\ 0 \\ 0.02120070 \end{bmatrix}$$

Local element end forces

Local end force order:

$$[F'_{xi}, F'_{yi}, M_i,\ F'_{xj}, F'_{yj}, M_j]$$

Use the corrected relationship:

$$\mathbf{Q}'^{(e)} = \mathbf{k}'_e \mathbf{u}'_e + \mathbf{Q}'^{(e)}_f$$

Element 1:

Local stiffness:

$$\mathbf{k}'_{e1} = \begin{bmatrix} 94800 & 0 & 0 & -94800 & 0 & 0 \\ 0 & 53.28 & 266.4 & 0 & -53.28 & 266.4 \\ 0 & 266.4 & 1776 & 0 & -266.4 & 888 \\ -94800 & 0 & 0 & 94800 & 0 & 0 \\ 0 & -53.28 & -266.4 & 0 & 53.28 & -266.4 \\ 0 & 266.4 & 888 & 0 & -266.4 & 1776 \end{bmatrix}$$

Local fixed-end vector for the horizontal UDL on the column (in local order):

$$\mathbf{Q}'^{(e1)}_f = \begin{bmatrix} 0 \\ 120 \\ 200 \\ 0 \\ 120 \\ -200 \end{bmatrix} \quad (kN, kN, kN \cdot m)$$

Therefore,

$$\mathbf{Q}'^{(e1)} = \mathbf{k}'_{e1} \mathbf{u}'_{e1} + \mathbf{Q}'^{(e1)}_f = \begin{bmatrix} 56.831 \\ 126.815 \\ 222.802 \\ -56.831 \\ 113.185 \\ -154.651 \end{bmatrix} \quad (kN, kN, kN \cdot m)$$

Element 2:

Local stiffness:

$$\mathbf{k}'_{e2} = \begin{bmatrix} 118500 & 0 & 0 & -118500 & 0 & 0 \\ 0 & 104.0625 & 416.25 & 0 & -104.0625 & 416.25 \\ 0 & 416.25 & 2220 & 0 & -416.25 & 1110 \\ -118500 & 0 & 0 & 118500 & 0 & 0 \\ 0 & -104.0625 & -416.25 & 0 & 104.0625 & -416.25 \\ 0 & 416.25 & 1110 & 0 & -416.25 & 2220 \end{bmatrix}$$

Local fixed-end vector for the midspan point load on the beam (local = global here):

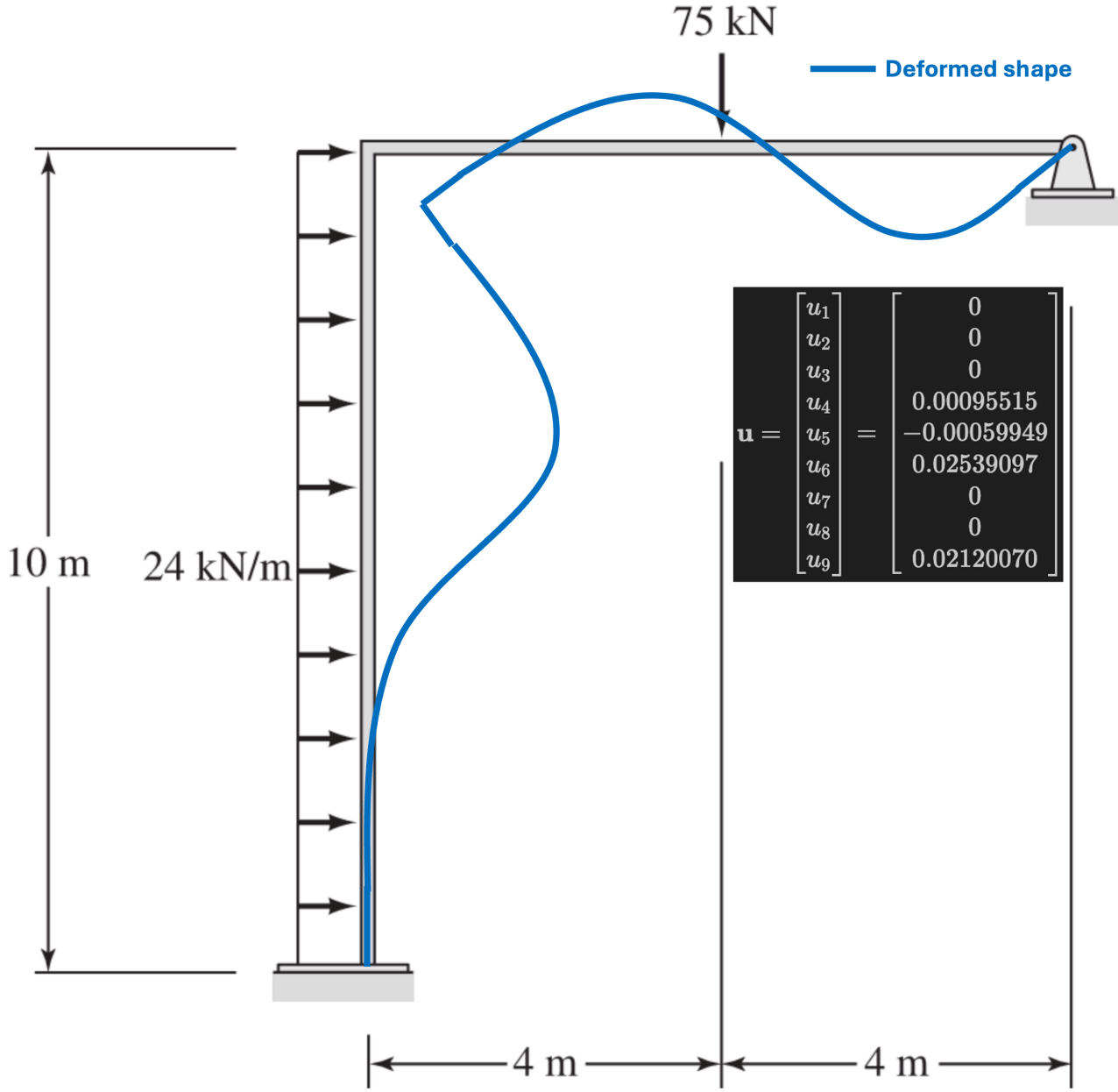
$$\mathbf{Q}_f'^{(e2)} = \begin{bmatrix} 0 \\ 37.5 \\ 75 \\ 0 \\ 37.5 \\ -75 \end{bmatrix} \quad (kN, kN, kN\cdot m)$$

Therefore,

$$\mathbf{Q}'^{(e2)} = \mathbf{k}'_{e2}\mathbf{u}'_{e2} + \mathbf{Q}_f'^{(e2)} = \begin{bmatrix} 113.185 \\ 56.831 \\ 154.651 \\ -113.185 \\ 18.169 \\ 0 \end{bmatrix} \quad (kN, kN, kN\cdot m)$$

Deformed shape

- Sketch a smooth deflection curve that bends upward/downward between spans consistent with these signs, returning to zero deflection at each support.



$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \\ u_9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.00095515 \\ -0.00059949 \\ 0.02539097 \\ 0 \\ 0 \\ 0.02120070 \end{bmatrix}$$